

6	(a) (i) movement/flow of charged particles	B1	[1]
	(ii) work done per unit charge (transferred)	B1	[1]
	(b) straight line through origin resistance = V/I , with values for V and I shown = $20\ \Omega$ (using the gradient loses the last mark)	B1 M1 A0	[2]
	(c) (i) 0.5 A	A1	[1]
	(ii) <i>either</i> resistance of each resistor is $20\ \Omega$ <i>or</i> total current = 0.8 A <i>either</i> combined resistance = $10\ \Omega$ <i>or</i> $R = E/I = 10\ \Omega$	C1 A1	[2]
	(d) (i) 10 V	A1	[1]
	(ii) power = EI = $10 \times 0.2 = 2.0\text{ W}$	C1 A1	[2]

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7 (a) (i) either helium nucleus